

每一层里到底装了什么 — Spec 清单

每层可落地、可构建的具体部件。对齐真实格式：Anthropic SKILL.md、MCP tools、Claude Code、OpenAI connectors。

L3 · Skills

SkillSpec — Anthropic SKILL.md 格式

- name + description (何时使用)
- input / output schema
- output format + 约束
- call / stop / loop 条件
- examples (shots) + 反例
- 个性化 + verification

L2 · Harness

CapabilityCapsule — 以 MCP 为底座

- CapabilityCapsule: inputSchema · contract · invariants
- 工具路由 (intent-scoped, top-k)
- context 组装 (shots · schema · memory)
- code generation (Read/Edit/Bash + test/lint)
- tool synthesis (沙箱验证的 capsule)
- 记忆维护 M (compaction · writeback
- · reflection · retrieval · GC)
- reward / verifier + policy gate + 审计日志

L1 · Scaffold

ExecutionSpec — 隔离与资源

- 隔离: microVM · container · wasm
- bash / process / filesystem
- 网络: allowlist · egress 策略
- SSO / 身份: OIDC · SAML · SCIM · RBAC
- 云: AWS/GCP/Azure · KMS · S3 · PrivateLink
- serving (KV cache · batching) · lifecycle · 观测

数据子系统 $\mathcal{D}$

- › D_1 DataSourceCard — source · auth · quota · NFR
- › D_2 SemanticSummary — schema* · relations · themes
- › D_3 DataUsageSkill — sources · join plan · runs
- › D_4 Lifetime — freshness · retention · budget
- › Ω 工作区 — RunTrace · ArtifactManifest · LLM Wiki